

SmartCapture Pro

UX Design Specification

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1. Introduction

1.1 Document Purpose

This UX Design Specification document serves as the definitive visual and interaction design reference for SmartCapture Pro, a Chrome extension designed to revolutionize the way users capture, annotate, and share web content. The document provides comprehensive guidance for all stakeholders involved in the development process, including frontend engineers, UI designers, QA testers, and product managers. Every visual element, interaction pattern, and component behavior described herein must be faithfully implemented to ensure a consistent and delightful user experience across all extension surfaces.

The specification covers the complete design language system, from foundational elements such as color palettes and typography scales to complex interaction flows like full-page screenshot capture, intelligent annotation tools, and visual diff comparison views. Each section includes detailed wireframe descriptions, component specifications, state definitions, and accessibility requirements. The design philosophy prioritizes minimal cognitive load, fast task completion, and seamless integration with the Chrome browser ecosystem, ensuring that users can accomplish their goals with maximum efficiency and minimum friction.

1.2 Design Philosophy

SmartCapture Pro follows a design philosophy rooted in three core principles that guide every visual and interaction decision throughout the extension. These principles ensure the product delivers a premium experience that feels native to Chrome while providing powerful capabilities that go far beyond the default browser screenshot functionality.

Instant Clarity: Every screen must communicate its purpose within 200 milliseconds of appearing. Users should never have to guess what action to take next. Visual hierarchy, clear labels, and contextual cues are used consistently to guide attention toward primary actions while keeping secondary options accessible but visually recessive. The information architecture is organized around user mental models rather than technical implementation details.

Progressive Disclosure: The extension reveals complexity gradually based on user intent. The default popup presents only the most common actions (capture full page, capture visible area, capture selection). Advanced features such as OCR text extraction, visual diff comparison, and annotation tools are accessible but never intrusive. This approach ensures that first-time users feel confident immediately while power users can discover and master advanced capabilities at their own pace.

Seamless Performance: Every interaction provides immediate visual feedback within 100 milliseconds. Animations are purposeful and brief, never exceeding 300 milliseconds for state transitions. The UI remains responsive even during computationally intensive operations like full-page scrolling and capture, ensuring that the capture process feels instantaneous rather than mechanical. Loading states, progress indicators, and skeleton screens prevent any moment of uncertainty.

1.3 Target Platforms & Constraints

The extension operates within Google Chrome using Manifest V3, which imposes specific constraints on the design system. All popup UI elements must render correctly within a fixed viewport of 350 pixels wide by up to 600 pixels tall, as dictated by the Chrome Extension popup API. Content script overlays inject UI elements directly into the host page DOM, requiring careful isolation through Shadow DOM to prevent style conflicts with arbitrary website stylesheets. The design system must account for these constraints while maintaining visual consistency across all surfaces.

Surface	Viewport Size	Context	Notes
Popup (Default)	350 x 600 px	Browser toolbar	Scrollable if content exceeds height
Popup (Expanded)	350 x 600 px	Browser toolbar	Tab-based navigation within popup
Content Overlay	Full viewport	Host page DOM	Shadow DOM isolated, z-index: 2147483647
Side Panel	300 x full height	Chrome side panel	Chrome 114+, resizable
Options Page	Full browser tab	Standalone page	Standard web page, no viewport limits
Notifications	350 x auto (max 80)	System notification	Chrome notification API

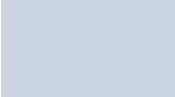
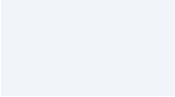
2. Design System Foundation

2.1 Color Palette

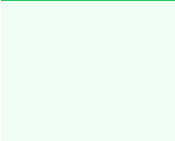
The SmartCapture Pro color system is built on a carefully calibrated palette that balances professionalism with visual appeal. The primary color, Deep Cyan (#0EA5E9), was selected for its strong association with technology, trust, and clarity. It provides excellent visibility against both light and dark backgrounds, ensuring that interactive elements and status indicators are

immediately recognizable. The palette includes semantic color assignments for success, warning, error, and informational states, each chosen to meet WCAG 2.1 AA contrast requirements when paired with their intended background colors.

2.1.1 Primary Color Tokens

Swatch	Color Name	Hex Code	Usage Context
	Primary (Deep Cyan)	#0EA5E9	Primary buttons, links, active states, brand identity
	Primary Dark	#0284C7	Button hover states, pressed states, emphasis
	Primary Light	#7DD3FC	Secondary backgrounds, selection highlights, subtle accents
	Primary Surface	#F0F9FF	Light tinted backgrounds for primary-themed sections
	Neutral 900	#0F172A	Primary text, headings, critical UI elements
	Neutral 700	#334155	Secondary text, descriptions, supporting content
	Neutral 500	#64748B	Placeholder text, disabled labels, tertiary content
	Neutral 300	#CBD5E1	Borders, dividers, inactive interactive elements
	Neutral 100	#F1F5F9	Card backgrounds, subtle section differentiation
	White	#FFFFFF	Primary background, content areas, input fields

2.1.2 Semantic Colors

Swatch	Color Name	Hex Code	Usage Context
	Success	#22C55E	Capture complete, save successful, OCR ready
	Success Surface	#F0FDF4	Success notification backgrounds, positive state cards
	Warning	#F59E0B	Storage nearly full, large file warnings, retry prompts

Swatch	Color Name	Hex Code	Usage Context
	Warning Surface	#FFFBEB	Warning notification backgrounds, caution state cards
	Error	#EF4444	Capture failure, permission denied, network errors
	Error Surface	#FEF2F2	Error notification backgrounds, failure state cards
	Info	#3B82F6	Tips, help text, informational tooltips, guidance
	Info Surface	#EFF6FF	Informational backgrounds, help panels, onboarding

2.2 Typography

The typographic system for SmartCapture Pro uses Inter as the primary typeface family, chosen for its exceptional legibility at small sizes, excellent screen rendering characteristics, and comprehensive weight range from 400 (Regular) through 700 (Bold). Inter was specifically designed for computer interfaces and provides clear differentiation between similar characters such as uppercase l, lowercase l, and the numeral 1, which is critical for a productivity tool where users frequently read technical content, URLs, and data values. The type scale follows a modular ratio of 1.25 (Major Third), providing a harmonious progression from caption text at 11px through large display headings at 24px within the popup context.

Token	Font	Size (px)	Weight	Line Height	Usage
display-lg	Inter	24	700	32px	Popup title, main headings
display-md	Inter	20	600	28px	Section headings, dialog titles
display-sm	Inter	18	600	24px	Subsection headings, card titles
body-lg	Inter	14	400	20px	Primary body text, descriptions
body-md	Inter	13	400	18px	Secondary text, list items
body-sm	Inter	12	400	16px	Captions, timestamps, metadata

Token	Font	Size (px)	Weight	Line Height	Usage
label	Inter	11	500	16px	Button labels, form labels, tags
code	JetBrains Mono	12	400	16px	URL display, file sizes, technical data

2.3 Spacing & Layout Grid

The spacing system uses a base unit of 4 pixels, with all spacing values being multiples of 4. This creates a consistent visual rhythm throughout the interface and ensures pixel-perfect alignment across all components. The grid system for the popup uses 4-column layout with 16px gutters, providing flexibility for different content arrangements while maintaining visual coherence. Content margins are set at 16px on all sides within the popup, with 20px padding within cards and 12px spacing between form elements. This consistent spacing framework ensures that regardless of screen density or user zoom level, the interface maintains its carefully crafted proportions and visual balance.

Token	Value	Common Usage
space-1	4px	Icon to label gap, inline element spacing
space-2	8px	Tight element spacing, related controls
space-3	12px	Form element vertical gap, list item padding
space-4	16px	Section internal padding, card padding
space-5	20px	Card internal padding, content grouping
space-6	24px	Section gaps, major content separation
space-8	32px	Major section breaks, vertical rhythm
space-10	40px	Page-level vertical spacing
popup-pad	16px	Popup content area margins
card-pad	20px	Card content internal padding

2.4 Icon System

SmartCapture Pro uses a consistent icon system based on 20x20 pixel SVG icons drawn on a 24x24 artboard with 2px padding. All icons use the `currentColor` CSS property, allowing them to automatically adapt to their context (inheriting text color for labels, or explicit color for standalone use). The icon weight is set to 1.5px stroke width with round line caps and round line joins, creating a friendly yet precise visual style that aligns with modern design trends in

productivity applications. Icons follow Material Design naming conventions where applicable, supplemented by custom icons for extension-specific features such as the capture reticle, stitch progress indicator, and diff comparison arrows.

The icon library is organized into four categories: Navigation Icons (arrow, chevron, close, menu), Action Icons (capture, crop, annotate, download, share, delete, copy), Status Icons (success checkmark, error X, warning triangle, info circle, loading spinner), and Feature Icons (OCR text scan, visual diff overlay, AI annotation wand, image compare). Each category maintains visual consistency within its group while allowing the four categories to be visually distinct at a glance, helping users quickly identify the type of action or information an icon represents.

3. Core UI Screens

3.1 Extension Popup (Main Menu)

The extension popup serves as the primary entry point for all SmartCapture Pro interactions. It opens when the user clicks the extension icon in the Chrome toolbar and presents a clean, focused interface that prioritizes the three most common capture actions. The popup uses a vertical layout with a branded header area at the top, followed by the main action buttons, a recent captures section, and a bottom toolbar for settings access and account information. The entire popup fits within the 350x600 pixel constraint without scrolling in its default state, ensuring instant access to core functionality.

3.1.1 Popup Layout Structure

Header Section [350 x 56px]

SmartCapture Pro logo (32x32) aligned left with app name 'SmartCapture Pro' in display-sm weight. Version badge 'v1.0' in label text positioned right-aligned. Bottom border: 1px Neutral-300. Total height: 56px.

Capture Actions Section [318 x 148px (within 16px padding)]

Three vertically stacked action buttons, each 40px tall with 12px vertical spacing. Button 1: Camera icon + 'Capture Full Page' (primary style, Deep Cyan bg). Button 2: Crop icon + 'Capture Visible Area' (secondary style, outline). Button 3: Selection icon + 'Capture Selection' (secondary style, outline). Each button uses rounded-lg (8px radius) with hover scale(1.02) transition.

Quick Tools Section [318 x 72px]

Horizontal row of 4 circular tool icons (36x36 each, 8px spacing). Tools: 'OCR Scan' (text icon), 'Visual Diff' (compare icon), 'Annotate' (pen icon), 'History' (clock icon). Each icon has a tooltip on hover. Section label 'Quick Tools' in label style above the row. Divider line above this section (1px Neutral-300).

Recent Captures Section [318 x 80px]

Horizontal scrollable row of capture thumbnails (64x48 each, 8px gap). Each thumbnail has a 2px rounded border that turns Primary on hover. Max 5 visible, scroll arrow indicators on edges. Label 'Recent' in label style with count badge. Timestamp below each thumbnail in body-sm. Empty state: dashed border box with 'No captures yet' in Neutral-500 centered text.

Footer Toolbar [350 x 48px]

Fixed to bottom of popup. Horizontal layout: Settings gear icon (left), Pro upgrade badge (center, accent background pill), User avatar circle (right, 28px). Background: Neutral-100 surface. Top border: 1px Neutral-300. Height: 48px. Settings icon navigates to Options page. Pro badge shows 'Upgrade to Pro' for free users or 'Pro' for subscribers.

3.1.2 Popup Interaction States

Element	Default State	Hover State	Active/Pressed State	Disabled State
Primary Button	bg: #0EA5E9, text: White	bg: #0284C7, scale(1.02), shadow-md	bg: #0369A1, scale(0.98)	bg: #CBD5E1, text: #94A3B8, no pointer
Secondary Button	border: 1.5px Neutral-300, text: Neutral-700	border: #0EA5E9, text: #0EA5E9, bg: Primary Surface	bg: #E0F2FE, border: #0284C7	border: #E2E8F0, text: #CBD5E1
Tool Icon	color: Neutral-500, bg: transparent	color: #0EA5E9, bg: Primary Surface, scale(1.1)	color: #0284C7, scale(0.95)	color: #E2E8F0
Thumbnail	border: 2px Neutral-200	border: 2px #0EA5E9, shadow-sm	border: 2px #0284C7	opacity: 0.5

3.2 Capture Progress Overlay

When a capture operation begins, a semi-transparent overlay appears on top of the host page content, providing clear visual feedback about the capture progress. This overlay uses a fixed

position with z-index set to 2147483647 to ensure it appears above all page content, and is injected via Shadow DOM to prevent style conflicts. The overlay design communicates three key pieces of information simultaneously: the current capture status (scrolling and capturing), the progress through the page (percentage and visual position indicator), and the ability to cancel the operation at any time. The overlay uses a frosted glass aesthetic achieved through a backdrop-filter blur of 8px combined with a semi-transparent dark background at 40% opacity.

The progress indicator consists of two components: a thin progress bar at the very top of the viewport (3px tall, full width, Primary color fill) that shows linear progress through the page, and a central status card that displays the current step name and a pulsing animation. The status card uses rounded-xl (12px) border radius with a dark semi-transparent background and white text, creating a floating appearance that works well against any page content. During capture, the page auto-scrolls smoothly at 100ms intervals per viewport height, and each captured section briefly flashes with a cyan overlay (15% opacity, 200ms duration) to provide visual confirmation that sections are being captured.

3.2.1 Progress Overlay Wireframe

Top Progress Bar [100vw x 3px]

Full viewport width, 3px tall, fixed to top. Background: Neutral-200. Fill: Primary gradient (left to right, #0EA5E9 to #38BDF8). Animated width from 0% to 100% based on scroll position. Smooth transition: width 200ms ease. Subtle glow effect (box-shadow: 0 0 8px rgba(14,165,233,0.4)) on the fill edge.

Central Status Card [280 x auto (approx 120px)]

Centered on screen, width: 280px. Background: rgba(15,23,42,0.85) with backdrop-filter: blur(8px). Rounded-xl (12px). Contains: Step icon (32x32, animated pulse) + Step label ('Capturing page...' in white body-lg) + Progress text ('Section 3 of 12' in Neutral-300 body-sm) + Cancel button (text button, 'Cancel' in Neutral-400, underline on hover). Entrance animation: fadeIn(200ms) + slideUp(100ms).

3.3 Screenshot Preview & Editor

After a capture completes, the full-page screenshot is presented in a dedicated preview overlay that provides both viewing and editing capabilities. This is the most feature-rich surface in the extension and serves as the central hub for all post-capture operations. The preview overlay takes over the entire browser viewport with a dark background (Neutral-900 at 95% opacity) to maximize the visual impact of the captured content and minimize distractions. The screenshot is displayed in a scrollable container with a subtle drop shadow, and a comprehensive toolbar provides access to annotation, OCR, export, and sharing functions.

The preview editor layout follows a standard image editing pattern with a top toolbar for file-level actions, a left sidebar for annotation tools, the main canvas area in the center, and a right panel

for contextual information and property editing. This layout maximizes the editing canvas area while keeping all tools accessible within a single click. The annotation system is powered by Fabric.js and provides a full range of drawing and markup tools including freehand pen, rectangles, ellipses, arrows, text boxes, highlights, and blur/redaction tools. Each annotation is stored as a separate Fabric.js object, allowing for independent selection, editing, deletion, and z-order management.

3.3.1 Preview Layout Wireframe

Top Action Bar [100vw x 48px]

Full viewport width, 48px tall, bg: #0F172A (Neutral-900). Left group: Back arrow icon + 'Back' label + filename ('capture-2026-04-29.png' in code font body-sm, Neutral-400). Center group: Undo/Redo button pair (outline style). Right group: Zoom controls (100% label with +/- buttons) + Download button (primary) + Share button (secondary) + Close button (X icon). Border-bottom: 1px rgba(255,255,255,0.1).

Left Annotation Toolbar [52px x calc(100vh - 48px)]

Vertical toolbar, 52px wide, full height minus top bar. Bg: #1E293B (Neutral-800). Contains 8 tool buttons (40x40 each, 6px vertical spacing): Select/Move (cursor icon), Pan/Hand (hand icon), Draw (pen icon), Rectangle (square icon), Ellipse (circle icon), Arrow (arrow-right icon), Text (T icon), Highlight (highlight icon). Active tool has Primary bg with white icon. Hover shows tool name tooltip to the right. Bottom: Eraser icon + Clear All button (with confirmation).

Main Canvas Area [calc(100vw - 52px) x calc(100vh - 48px)]

Remaining viewport space (calc(100vw - 52px) x calc(100vh - 48px)). Dark background: #0F172A. Screenshot centered with subtle shadow (0 4px 24px rgba(0,0,0,0.4)). Scrollable vertically with smooth scroll behavior. Zoom range: 25% to 400% with scroll-wheel zoom (Ctrl+scroll). Fabric.js canvas overlay for annotations. Cursor changes based on active tool (crosshair for draw, move for select, text cursor for text).

Right Properties Panel (Contextual) [240px x calc(100vh - 48px)]

240px wide panel, slides in from right when annotation is selected. Bg: #1E293B. Header: 'Properties' in display-sm + Close panel button. Content varies by selection: For shapes - fill color, border color, border width, opacity sliders. For text - font family dropdown, font size, bold/italic/underline toggles, text color, text alignment. For highlights - color picker (Yellow, Green, Pink, Blue, Orange presets). Bottom: 'Delete Annotation' button (destructive style, red).

3.3.2 Annotation Tool Specifications

Tool	Cursor	Default Style	Keyboard Shortcut	Behavior
Select/Move	default	N/A	V	Click to select, drag to move, resize handles on corners
Pan/Hand	grab / grabbing	N/A	H	Drag to pan canvas, scroll-wheel also pans
Draw (Freehand)	crosshair	stroke: #EF4444, width: 2px	D	Click+drag to draw freehand path, smooth on release
Rectangle	crosshair	stroke: #EF4444, fill: transparent, width: 2px	R	Click+drag from corner to corner, hold Shift for square
Ellipse	crosshair	stroke: #EF4444, fill: transparent, width: 2px	E	Click+drag from center outward, hold Shift for circle
Arrow	crosshair	stroke: #EF4444, width: 2px, headSize: 12	A	Click start, drag to end point, rotates freely
Text	text	font: Inter 14px, color: #0F172A	T	Click to place text box, auto-focus, Escape to confirm
Highlight	crosshair	fill: #FBBF24, opacity: 0.3	L	Click+drag rectangular highlight area, semi-transparent fill

3.4 OCR Results Panel

The OCR (Optical Character Recognition) Results Panel provides a dedicated interface for viewing, selecting, and copying text extracted from captured screenshots. Powered by Tesseract.js running entirely within the browser, the OCR engine processes the captured image locally without requiring any server-side computation, ensuring user privacy and zero-latency operation. The panel appears as a slide-in overlay from the right side of the preview editor, occupying 360 pixels of width and the full height of the viewport. This positioning allows users to simultaneously view the original screenshot and the extracted text, enabling easy verification and correction of OCR results.

The panel interface is divided into three functional zones: a compact header with processing status and action controls, a scrollable text content area that displays the extracted text with paragraph-level confidence indicators, and a bottom action bar for bulk operations. Each

paragraph of extracted text is displayed in a separate card with a confidence score badge (High/Medium/Low) and a visual indicator showing its corresponding region on the original screenshot. Clicking a paragraph highlights both the text and its corresponding region on the screenshot canvas, creating a direct visual link between the extracted content and its source. Users can edit any extracted text inline, and all edits are preserved even when the panel is closed and reopened.

3.4.1 OCR Panel Wireframe

OCR Header Bar [360 x 48px]

Full panel width (360px), 48px tall. Bg: #1E293B. Left: Back arrow icon + 'OCR Results' in display-sm. Center: Status badge ('Processing...' in warning color / 'Complete' in success color / '3 regions found' in info color). Right: Select All checkbox + Copy All button (primary mini). Progress bar (2px) below header shows OCR processing progress. Animates during processing, solid green when complete.

Text Content Area [360 x calc(100vh - 96px)]

Full panel width, scrollable remaining height. Bg: #0F172A. Contains individual paragraph cards stacked vertically (12px gap). Each card: 8px left border color-coded by confidence (green=High >90%, yellow=Medium 70-90%, red=Low <70%), text content in body-lg (Neutral-200), confidence percentage in body-sm (Neutral-500), timestamp of capture. Click to select, Shift+click for multi-select. Selected cards show Primary border highlight. Edit icon appears on hover.

OCR Bottom Action Bar [360 x 48px]

Full panel width, 48px tall. Bg: #1E293B. Top border: 1px rgba(255,255,255,0.1). Left: 'Copy Selected' button (primary mini, disabled when nothing selected). Center: Selection count ('3 paragraphs selected' in body-sm). Right: 'Export as TXT' dropdown button (secondary mini). All actions provide toast confirmation on completion.

3.5 Visual Diff Comparison View

The Visual Diff Comparison View enables users to compare two screenshots side by side with pixel-level difference highlighting, making it an invaluable tool for QA testers, designers reviewing design iterations, and professionals tracking changes in web content over time. The comparison engine uses the Pixelmatch library to compute differences between two images at the pixel level, generating a diff overlay that highlights changed regions in distinct colors while leaving unchanged regions transparent. The diff view supports three comparison modes: Side-by-Side (both images visible simultaneously with a synced scroll), Overlay (images stacked with diff highlights), and Onionskin (adjustable opacity blend of both images).

The diff interface provides intuitive controls for navigating between detected changes, including Previous/Next change navigation buttons and a change list panel that summarizes all detected differences with thumbnail previews and confidence percentages. Each detected change region is numbered and listed in the sidebar, allowing users to jump directly to specific changes. The color coding for diff regions follows a standard convention: red highlights for deletions (content present in the base image but missing in the comparison image), green highlights for additions (new content in the comparison image), and yellow highlights for modifications (content that exists in both but has changed). Users can adjust the diff sensitivity threshold using a slider control, with higher sensitivity detecting more subtle changes but also potentially more false positives from anti-aliasing artifacts and compression differences.

3.5.1 Diff View Layout

Diff Control Bar [100vw x 48px]

Full viewport width, 48px tall, bg: #0F172A. Left: Back arrow + 'Visual Diff' label. Center: Mode toggle (3 segmented buttons): 'Side by Side' / 'Overlay' / 'Onionskin'. Active mode has Primary bg. Right: Sensitivity slider (range: 1-100, default: 50) with label 'Sensitivity' + Change count badge ('12 changes detected'). Below: thin change navigation bar showing change distribution across image height.

Side-by-Side Panels [100vw x calc(100vh - 96px)]

Two equal-width panels (each calc(50vw - 100px)), full remaining height. Left panel: 'Before' image with cyan border-top (2px). Right panel: 'After' image with orange border-top (2px). Both panels scroll synchronously. Change regions highlighted with colored semi-transparent overlays (red/green/yellow). Clicking a change region in either panel selects it and shows details. A draggable divider between panels allows adjusting relative width (20%-80% range).

Change List Sidebar [280px x calc(100vh - 96px)]

280px wide, slides in from right on toggle. Bg: #1E293B. Header: 'Changes' label + count badge. Scrollable list of change items, each showing: change number, type icon (added/removed/modified), thumbnail (40x30), position label ('Top', 'Middle', 'Bottom'), confidence %. Active change has Primary left border. Click navigates both panels to that change position. Bottom: 'Export Diff Report' button.

4. Component Library

4.1 Button System

The button system provides a comprehensive set of interactive button variants designed to cover all interaction scenarios within the extension. Every button variant includes five distinct visual states: default, hover, active/pressed, focus (keyboard navigation), and disabled. The system uses

consistent border-radius (8px for standard, 20px for pill), padding (8px 16px for standard, 12px 24px for large), and transition timing (150ms ease for color transitions, 100ms ease for transform). Buttons use the Inter typeface at 500 weight and 14px size, with 16px line height and center alignment. Icon buttons use a 36x36px touch target with the icon centered, maintaining adequate click area even in the constrained popup environment.

Variant	Background	Border	Text Color	Use Case
Primary	#0EA5E9	none	#FFFFFF	Main CTA, capture actions, confirm
Primary Hover	#0284C7	none	#FFFFFF	Hover state for primary
Secondary	transparent	1.5px #CBD5E1	#334155	Secondary actions, cancel, go back
Ghost	transparent	none	#64748B	Tertiary actions, inline controls
Danger	#EF4444	none	#FFFFFF	Delete, clear, destructive actions
Danger Outline	transparent	1.5px #EF4444	#EF4444	Secondary destructive action
Success	#22C55E	none	#FFFFFF	Save, complete, positive confirm
Icon Button	transparent	1px transparent	#64748B	Toolbar buttons, settings icons
Icon Button Active	#0EA5E9 (10% opacity bg)	none	#0EA5E9	Selected tool/active state
Disabled (any)	#F1F5F9	none / 1px #E2E8F0	#94A3B8	Unavailable actions

4.2 Form Controls

Form controls within the extension follow a consistent design language that ensures visual harmony across settings pages, filter dialogs, and annotation property panels. All form inputs use a 36px height with 8px border-radius, 12px horizontal padding, and a 1.5px border in Neutral-300 that transitions to Primary color on focus. The focus state includes a subtle blue ring (box-shadow: 0 0 3px rgba(14,165,233,0.15)) that provides clear visual feedback for keyboard navigation and accessibility compliance. Placeholder text uses Neutral-500 at 13px size, and all inputs support both dark and light theme variants through CSS custom properties.

Select dropdowns use a custom-styled appearance with a chevron-down indicator icon positioned at the right edge, replacing the native browser dropdown styling for visual consistency. Checkboxes and radio buttons use custom SVG icons that animate smoothly between checked and unchecked states with a 150ms transition. Toggle switches use a 36x20px pill shape with a 16px circular knob that slides horizontally when toggled, using the Primary color for the active state and Neutral-300 for the inactive state. All form controls include proper ARIA attributes, visible focus indicators, and label associations to ensure WCAG 2.1 AA compliance.

4.3 Toast Notifications

The toast notification system provides non-intrusive feedback for user actions, system events, and status updates. Toasts appear at the bottom-center of the viewport (or bottom of the popup for popup-context notifications) with a smooth slide-up animation (200ms ease-out) and auto-dismiss after 4 seconds unless the user hovers over them (which pauses the auto-dismiss timer). Each toast variant uses a distinct color scheme: success (green), error (red), warning (amber), and info (blue), with corresponding icons and surface colors. Toasts support optional action buttons for inline resolution of issues (for example, a 'Retry' button on a failed capture toast) and a manual dismiss button (X icon) on the right side.

Variant	Background	Icon	Border Left	Auto-Dismiss
Success	#F0FDF4	Checkmark (green)	3px #22C55E	3 seconds
Error	#FEF2F2	X circle (red)	3px #EF4444	5 seconds (persistent option)
Warning	#FFFBEF	Triangle (amber)	3px #F59E0B	4 seconds
Info	#EFF6FF	Info circle (blue)	3px #3B82F6	3 seconds

4.4 Loading & Progress Indicators

The loading system uses three distinct indicator types, each designed for specific use cases and durations. For operations under 300 milliseconds, a subtle cursor change (to 'wait') is sufficient and no additional indicator is needed. For operations between 300ms and 2 seconds, a skeleton screen is displayed, which shows placeholder shapes matching the expected content layout with a gentle shimmer animation (linear gradient sweep from left to right, 1.5s infinite loop). For operations exceeding 2 seconds (such as full-page capture, OCR processing, and image diff computation), a determinate progress indicator is shown with percentage text, estimated time remaining, and a cancel button.

The skeleton screen shimmer uses a CSS animation that sweeps a semi-transparent white highlight (rgba(255,255,255,0.4)) diagonally across gray placeholder shapes (#E2E8F0) from left to right over 1.5 seconds, creating a premium loading feel. The spinner component is a 24x24px SVG

circle with a dashed stroke that rotates continuously (1s linear infinite), available in four colors matching the semantic palette (Primary, Success, Warning, Error). The linear progress bar fills from left to right with the Primary color gradient, accompanied by a subtle pulsing glow effect on the leading edge to draw attention to progress advancement.

5. Interaction Patterns & Micro-interactions

5.1 Hover & Focus States

Every interactive element in the extension provides clear visual feedback for both hover (mouse) and focus (keyboard) states. Hover transitions use a 150ms ease timing function, which is fast enough to feel responsive but slow enough to be perceptible, creating a polished interaction feel. Focus indicators use a 3px offset ring (box-shadow: 0 0 0 3px rgba(14,165,233,0.15)) that provides sufficient contrast for visibility while maintaining visual elegance. The focus ring appears only during keyboard navigation (using the :focus-visible CSS pseudo-class) and is suppressed for mouse clicks, preventing visual clutter for mouse users while maintaining full accessibility for keyboard and screen reader users.

Interactive elements with hover effects include buttons (color shift, scale transform, shadow elevation change), tool icons (color shift, subtle scale, tooltip appearance), thumbnail cards (border highlight, shadow elevation), text links (underline appearance, color shift), and navigation items (background highlight, left accent bar). Each hover effect is carefully calibrated to be noticeable without being distracting, using color changes rather than size changes as the primary signal. Scale transforms are limited to a maximum of 1.05x to prevent jarring movement, and shadow elevation changes use a single-step increase (shadow-sm to shadow-md) rather than multi-step elevation systems.

5.2 Animation & Transition Guidelines

All animations within the extension follow a consistent timing and easing framework designed to feel natural and responsive. The standard easing function for UI transitions is cubic-bezier(0.4, 0, 0.2, 1), which provides a fast start that decelerates smoothly, matching the natural physics of objects in motion. Exit animations use a faster timing (100ms) compared to entrance animations (200ms), following the principle that users perceive leaving objects as less important than arriving ones. All animations respect the prefers-reduced-motion media query, with motion-sensitive users receiving instant state changes instead of animated transitions.

Interaction	Duration	Easing	Property	Notes
Button hover	150ms	ease	background-color, transform	No delay, immediate feedback
Panel slide-in	200ms	cubic-bezier(0.4, 0, 0.2, 1)	transform: translateX	From right edge, slight overshoot
Panel slide-out	100ms	cubic-bezier(0.4, 0, 1, 1)	transform: translateX	Faster than entrance
Toast enter	200ms	ease-out	transform: translateY, opacity	Slide up from below
Toast exit	150ms	ease-in	transform: translateY, opacity	Slide down to below
Progress bar fill	500ms	linear	width	Smooth continuous fill
Skeleton shimmer	1500ms	linear	background-position	Infinite loop, left to right sweep
Page capture flash	200ms	ease-out	opacity	15% cyan overlay, single flash per section
Modal backdrop	200ms	ease-out	opacity	0 to 0.5 dark overlay
Modal content	200ms	cubic-bezier(0.4, 0, 0.2, 1)	transform: scale(0.95 to 1), opacity	Subtle scale-up entrance
Zoom change	150ms	ease-out	transform: scale	Ctrl+scroll in preview canvas
Tooltip show	100ms	ease-out	opacity, transform: translateY(-4px)	8px offset from trigger element

5.3 Keyboard Shortcuts

The extension provides a comprehensive set of keyboard shortcuts designed for power users who prefer keyboard-driven workflows. All shortcuts use the Alt key as the primary modifier to avoid conflicts with website-specific shortcuts and Chrome's built-in keyboard commands. Shortcuts are available in both the popup context and the preview editor context, with context-specific bindings displayed in tooltip text on hover. The shortcut system uses a visual shortcut hint pattern where available shortcuts are displayed as small gray key badges (for example, 'Alt+S') next to their corresponding menu items and buttons.

Shortcut	Context	Action	Notes
Alt+Shift+S	Global (any page)	Capture Full Page	Opens extension popup if needed, starts capture
Alt+Shift+V	Global (any page)	Capture Visible Area	Quick capture of current viewport
Alt+Shift+D	Preview Editor	Open Visual Diff	Requires at least one previous capture
Alt+Shift+O	Preview Editor	Run OCR	Starts OCR processing on current screenshot
Escape	Any overlay	Close overlay / Cancel	Closes panels, modals, exits editor
Ctrl+Z	Preview Editor	Undo annotation	Undo last annotation action
Ctrl+Shift+Z	Preview Editor	Redo annotation	Redo last undone annotation
Ctrl+S	Preview Editor	Save / Download	Downloads annotated screenshot
Ctrl+C	Preview Editor	Copy to clipboard	Copies screenshot to system clipboard
Ctrl+= / Ctrl+-	Preview Editor	Zoom in / out	Adjusts canvas zoom level
Ctrl+0	Preview Editor	Reset zoom to 100%	Returns to default zoom level
V	Preview Editor	Select/Move tool	Switches to selection tool
D	Preview Editor	Draw tool	Switches to freehand drawing

Shortcut	Context	Action	Notes
R	Preview Editor	Rectangle tool	Switches to rectangle annotation
Delete / Backspace	Preview Editor	Delete selected	Removes selected annotation

6. Dark & Light Theme Specification

6.1 Theme Architecture

SmartCapture Pro implements a dual-theme system using CSS custom properties (CSS variables) that allows instant switching between dark and light modes without page reload. The default theme is Dark, chosen to match the Chrome browser's native dark mode aesthetic and to minimize eye strain during extended screenshot review sessions. Users can switch themes via the Settings page, and the system also respects the user's operating system preference through the prefers-color-scheme media query when no explicit theme selection has been made. Theme transitions use a 200ms crossfade animation on all color properties to create a smooth visual transition when switching.

6.1.1 Dark Theme (Default)

Token	Value	Applied To
--bg-primary	#0F172A	Page backgrounds, overlay backgrounds
--bg-secondary	#1E293B	Cards, panels, toolbar backgrounds
--bg-tertiary	#334155	Input fields, hovered items
--text-primary	#F8F AFC	Headings, primary text, button text
--text-secondary	#94A3B8	Descriptions, captions, secondary content
--text-tertiary	#64748B	Placeholders, disabled labels
--border-primary	rgba(255,255,255,0.1)	Primary dividers, card borders
--border-secondary	rgba(255,255,255,0.05)	Subtle dividers, input borders
--accent	#0EA5E9	Primary interactive elements, links
--accent-hover	#38BDF8	Hover state for accent elements
--shadow	0 4px 24px rgba(0,0,0,0.4)	Card shadows, floating elements

6.1.2 Light Theme

Token	Value	Applied To
--bg-primary	#FFFFFF	Page backgrounds, overlay backgrounds
--bg-secondary	#F8FAFC	Cards, panels, toolbar backgrounds
--bg-tertiary	#F1F5F9	Input fields, hovered items
--text-primary	#0F172A	Headings, primary text, button text
--text-secondary	#475569	Descriptions, captions, secondary content
--text-tertiary	#94A3B8	Placeholders, disabled labels
--border-primary	#E2E8F0	Primary dividers, card borders
--border-secondary	#F1F5F9	Subtle dividers, input borders
--accent	#0284C7	Primary interactive elements, links
--accent-hover	#0EA5E9	Hover state for accent elements
--shadow	0 4px 12px rgba(0,0,0,0.08)	Card shadows, floating elements

7. Accessibility Guidelines

7.1 WCAG 2.1 AA Compliance

SmartCapture Pro is designed to meet WCAG 2.1 Level AA accessibility standards across all extension surfaces. This commitment ensures that users with visual, motor, auditory, or cognitive disabilities can effectively use all extension features. Color contrast ratios for all text-background combinations meet or exceed the 4.5:1 minimum for normal text and 3:1 for large text (18px+ or 14px+ bold). All interactive elements have a minimum touch target size of 36x36 pixels (exceeding the WCAG recommendation of 24x24px), ensuring comfortable interaction for users with motor impairments. The extension provides full keyboard navigation support with visible focus indicators, logical tab order, and keyboard shortcuts for all primary actions.

Screen reader compatibility is achieved through proper ARIA labeling of all interactive elements, live regions for dynamic content updates (such as capture progress and OCR status), and descriptive alt text for all non-decorative images. The extension uses semantic HTML elements where applicable (button, nav, main, dialog) and supplements with ARIA roles, states, and properties as needed for custom components. All form inputs have associated labels, error messages are linked to their respective fields via aria-describedby, and modal dialogs trap focus

appropriately. The color system never relies on color alone to convey information; all status indicators include supplementary text or icon differentiation.

7.2 Motion & Animation Preferences

The extension respects the prefers-reduced-motion media query at three levels. For users with prefers-reduced-motion: reduce, all non-essential animations are disabled, including skeleton shimmer effects, panel slide transitions, and decorative micro-interactions. Essential state feedback (such as progress bar advancement and toast notifications) uses instant opacity changes instead of animated transitions. Hover effects use only color changes without scale transforms. For users with prefers-reduced-motion: no-preference (the default), all standard animations play at their specified durations and easings. A custom accessibility setting in the extension options allows users to further customize animation behavior independently of their system preference, providing granular control over motion sensitivity.

8. Responsive & Adaptive Design

8.1 Popup Viewport Management

The extension popup presents unique responsive challenges due to Chrome's strict 350px width constraint and the 600px maximum height before scrolling is required. The layout system uses a flexbox-based approach with overflow management to ensure that the most critical actions are always visible without scrolling. The capture action buttons, which represent the primary use case, are always positioned in the upper portion of the popup above the fold. The recent captures section and quick tools section are designed to gracefully compress or collapse based on available vertical space. When the popup content exceeds 600px, a custom scrollbar with a 4px width and Primary color thumb appears, maintaining visual consistency with the design system.

The popup uses a dynamic content height calculation system that measures the total content height on load and on any state change (such as adding new recent captures or expanding a section) and adjusts the layout accordingly. If the viewport is constrained, the Recent Captures section switches from a horizontal scrollable row to a compact list view showing only the two most recent captures with a 'View All' link. The Quick Tools section can collapse into a smaller icon-only row if vertical space is limited. This adaptive behavior ensures that the extension remains usable across different Chrome window sizes, display densities, and zoom levels without requiring manual resizing by the user.

8.2 High-DPI Display Support

All graphical assets, icons, and canvas rendering within the extension are optimized for high-DPI (Retina) displays. SVG icons are used throughout the interface, providing infinite resolution scaling

without file size overhead. For bitmap-based operations such as screenshot capture and Fabric.js annotation canvas, the system detects the device pixel ratio (`window.devicePixelRatio`) and scales the canvas backing store accordingly, ensuring that captured and annotated screenshots maintain full resolution on high-DPI displays. The screenshot capture engine captures at the logical pixel resolution multiplied by the device pixel ratio, producing images that are physically larger but correctly represent the visual content at native display resolution. This ensures that captured screenshots look crisp when viewed on any display, including Retina MacBook screens and high-DPI Windows displays with 150% or 200% scaling.

9. Icon Library Reference

The complete icon library for SmartCapture Pro consists of 36 custom-designed SVG icons organized into four functional categories. Each icon is designed on a 24x24 pixel artboard with 2px padding, resulting in a 20x20 pixel visual area. The stroke weight is 1.5px with round line caps (`stroke-linecap: round`) and round line joins (`stroke-linejoin: round`), creating a consistent visual weight that works well at both the standard 20px display size and the 16px compact size used in dense toolbar layouts. All icons use the `currentColor` CSS property for automatic color adaptation, and each icon has been tested for clarity at both 1x and 2x display densities.

Category	Icon Name	Visual Description	Usage Context
Navigation	arrow-left	Left-pointing chevron arrow	Back navigation, return to previous
Navigation	arrow-right	Right-pointing chevron arrow	Forward navigation, expand section
Navigation	chevron-down	Downward-pointing chevron	Dropdown indicator, collapse section
Navigation	close	X shape with 45-degree rotation	Close panel, dismiss modal, cancel
Navigation	menu	Three horizontal lines (hamburger)	Overflow menu, additional options
Action	camera	Camera body with lens circle	Capture full page
Action	crop	Overlapping L-shaped corners	Crop tool, capture visible area
Action	selection	Dashed rectangle with crosshair	Capture selection tool
Action	download	Downward arrow with tray	Save/download screenshot
Action	share	Three connected dots with arrows	Share screenshot
Action	copy	Two overlapping rectangles	Copy to clipboard

Category	Icon Name	Visual Description	Usage Context
Action	delete	Trash can outline	Delete capture, remove annotation
Action	undo	Curved left arrow (counterclockwise)	Undo last action
Action	redo	Curved right arrow (clockwise)	Redo undone action
Annotation	cursor	Mouse pointer arrow	Select/move annotation tool
Annotation	hand	Open hand outline	Pan canvas tool
Annotation	pen	Diagonal pen/pencil	Freehand drawing tool
Annotation	square	Outlined square	Rectangle annotation tool
Annotation	circle	Outlined circle	Ellipse annotation tool
Annotation	arrow-right	Line with arrowhead	Arrow annotation tool
Annotation	type	Capital letter T	Text annotation tool
Annotation	highlight	Marker pen with highlight bar	Highlight annotation tool
Annotation	eraser	Eraser outline	Erase annotation tool
Feature	scan-text	Document with text lines + magnifier	OCR text scan feature
Feature	git-compare	Two overlapping circles	Visual diff comparison
Feature	wand-sparkles	Magic wand with sparkles	AI auto-annotation
Feature	history	Clock with counterclockwise arrow	Capture history
Status	check-circle	Circle with checkmark	Success state
Status	x-circle	Circle with X mark	Error state
Status	alert-triangle	Triangle with exclamation mark	Warning state
Status	info	Circle with letter i	Information state
Status	loader	Rotating dashed circle	Loading/processing state
General	settings	Gear/cog outline	Settings, configuration
General	user	Person bust outline	User profile, account
General	moon	Crescent moon	Dark mode toggle
General	sun	Circle with rays	Light mode toggle

10. User Flow Diagrams

10.1 Full Page Capture Flow

The full page capture flow represents the primary user journey, from initiating a capture through the popup to downloading or sharing the final annotated screenshot. This flow has been optimized to minimize the number of steps required for the most common use case (capture and download) while providing natural entry points for annotation, OCR, and diff comparison at each stage. The flow supports multiple exit points, allowing users to abandon the process at any stage without losing previously captured data, and includes robust error recovery paths for common failure scenarios such as permission denial, page navigation during capture, and insufficient storage space.

Step	User Action	System Response	UI State Change
1	Click extension icon in toolbar	Popup opens with main menu	Popup visible, focus on primary button
2	Click 'Capture Full Page'	Popup closes, overlay appears	Progress overlay with 'Starting capture...' status
3	Page auto-scrolls (automated)	Each viewport section captured	Progress bar advances, section flash indicators
4	All sections captured & stitched	Stitching animation (1s), then preview opens	Preview editor with full screenshot in canvas
5a	Click Download button	Image saved to Downloads folder	Toast: 'Screenshot saved successfully'
5b	Select annotation tool	Tool activated, canvas ready	Left toolbar highlights active tool, cursor changes
5c	Click OCR button in toolbar	OCR processing begins	Progress bar, 'Processing OCR...' status
6	Edit/annotate as needed	Fabric.js canvas interactive	Properties panel shows on annotation selection
7	Click Save/Share/Close	Action executed or editor closes	Return to host page, toast confirmation

10.2 Visual Diff Flow

The visual diff flow enables users to compare two screenshots and identify pixel-level differences between them. This flow supports three entry points: from the popup Quick Tools section (requiring selection of two captures from the history panel), from within the preview editor (comparing the current screenshot with a previously saved one), or from the context menu option (capturing a new screenshot specifically for comparison). The flow is designed to surface

actionable insights quickly, with the most critical changes highlighted first and navigation controls that allow efficient review of all detected differences.

Step	User Action	System Response	UI State Change
1	Click 'Visual Diff' in Quick Tools	Diff image selector panel opens	Modal showing recent captures grid
2	Select 'Before' image (base)	Image highlighted, border changes	Selected image shows cyan border
3	Select 'After' image (comparison)	Diff processing begins	Loading skeleton, 'Computing differences...'
4	Diff computation complete	Diff view opens with highlighted changes	Side-by-side panels with change overlays
5	Navigate changes with arrows	Viewport scrolls to change region	Change panel highlights current change item
6	Click 'Export Diff Report'	Report generated (PNG annotated)	Toast: 'Diff report saved', download initiated
7	Close diff view (Escape or X)	Return to popup or host page	Smooth exit transition, state cleaned up

11. Error & Empty States

11.1 Error State Design

Error states in SmartCapture Pro are designed to be informative, actionable, and emotionally neutral. Every error includes three components: a clear explanation of what went wrong (in plain language, never technical jargon), a suggested action to resolve the issue, and a visual indicator (icon and color) that immediately communicates the severity of the error. Error states use the Error color (red) as an accent rather than flooding the screen with red, preventing the visual anxiety that can result from large areas of alarming color. The error icon is always accompanied by a headline in body-lg weight and a description in body-md with secondary color, maintaining clear visual hierarchy.

Error Scenario	Headline	Description	Action Button	Icon
Permission denied	Camera permission required	Please grant screen capture permission in your browser settings to continue.	Open Settings	shield-x

Error Scenario	Headline	Description	Action Button	Icon
Page navigated during capture	Capture interrupted	The page changed during capture. The partial capture has been saved.	View Partial Capture	alert-triangle
Storage full (IndexedDB quota)	Storage almost full	You have less than 50MB of storage remaining. Please delete old captures.	Manage Storage	database
OCR failed	Text extraction failed	Could not extract text from this image. Try a clearer screenshot.	Retry OCR	scan-text
Image too large for diff	Image too large	The selected images exceed the maximum size for comparison (40MB each).	Choose Smaller Image	image
Network error (future cloud)	Connection lost	Could not save to cloud storage. Your capture is saved locally.	Save Locally	wifi-off
Invalid selection area	Selection too small	Please select a larger area (minimum 32x32 pixels).	Try Again	selection

11.2 Empty State Design

Empty states serve as both informative placeholders and gentle onboarding moments that guide users toward their first interaction with the extension. Each empty state includes a friendly illustration or icon (rendered in Neutral-300), a concise headline explaining what content would appear in this section, a brief description of how to populate it, and a prominent call-to-action button that leads the user to the relevant feature. The design avoids making empty states feel like errors; instead, they are framed as opportunities. For example, the empty Recent Captures section shows a camera icon with dashed border and the message 'Take your first screenshot' alongside a 'Capture Now' button, turning an absence of content into an invitation to engage with the product.

Location	Headline	Description	CTA Button
Recent Captures (popup)	No captures yet	Your screenshots will appear here after you capture a page.	Capture Full Page
Visual Diff Selector	Not enough captures	You need at least 2 captures to compare. Go capture some pages first.	Go Capture

Location	Headline	Description	CTA Button
OCR Results	No text detected	OCR could not find any text in this screenshot. Try a different image.	Close
Capture History	History is empty	All your past captures will be listed here for easy access.	Start Capturing
Annotation Layer	No annotations	Use the toolbar on the left to add annotations to this screenshot.	Select a Tool